Using Yelp Data to Estimate Business Demand & Identify Key Drivers for Location-Based Decision Support

Sunghoon Ahn • Eun Soo Park • Thiri Htet Oo • Jayanthi Muthuselvan | Master of Data Analytics Capstone | Prof. Touraj BaniRostam

DAMO-699-1 SPRING 2026

UNIVERSITY OF NIAGARA FALLS CANADA

PROBLEM & OBJECTIVE 1

Local businesses need scalable methods to estimate customer demand from digital engagement signals.

Research Questions

RQ1: How accurately can Yelp data estimate business demand?

RQ2: How do location and temporal factors influence demand?

RQ3: Which model performs best?

RQ4: Which features drive predicted demand?

Objective: predict check-in-based demand and explain key drivers for location-based decision support.

Target variable

$y = \log(\text{checkin_count} + 1)$

- Business demand is represented by log-transformed Yelp check-in activity.
- This avoids claiming real sales or POS transactions.
- Poster values are aligned to Notebook 03 and Notebook 04 outputs.

DATA & FEATURE MATRIX 2

Final feature table: **52,286** rows and **42** columns.

Modeling matrix: **52,286** businesses and **34** features.

Test set: **10,458** businesses

Feature groups

- Review activity and engagement
- Temporal activity and business hours
- Sentiment ratios and average sentiment
- Business attributes and nearby density

Data source

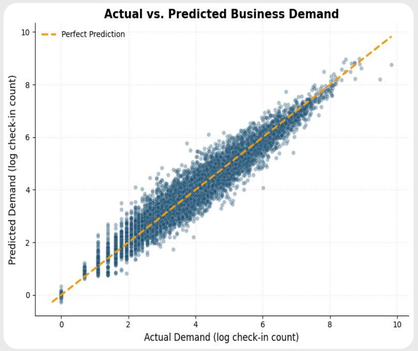
- Yelp Open Dataset, April 2024 version.
- Integrated business, review, and check-in information.

METHODOLOGY PIPELINE 3

01 INGEST → **02** FEATURE → **03** MODEL → **04** SHAP

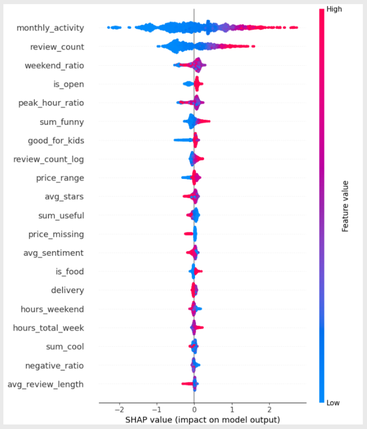
- Notebook 03: trains and evaluates baseline and ensemble models.
- Notebook 04: loads saved outputs, visualizes model behavior, and applies SHAP.
- Interpretation focuses on global and grouped feature effects, not causal proof.

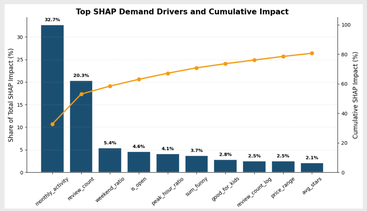
MODEL DIAGNOSTIC PLOT 4



- Actual-vs-predicted plot comes from Notebook 04.
- It supports the high R^2 result reported in the benchmark table.

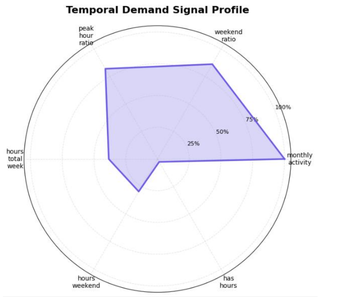
SHAP INTERPRETATION 5



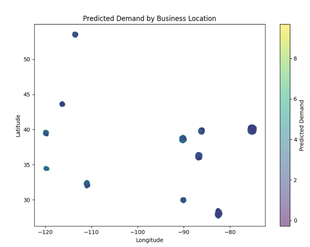


SHAP identifies monthly activity and review count as the strongest contributors, followed by weekend ratio, business status, and peak-hour ratio.

TEMPORAL & SPATIAL SIGNALS 6



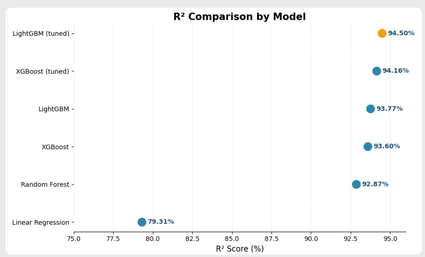
The radar chart shows a normalized temporal signal profile. Monthly activity, weekend behavior, and peak-hour ratio were the strongest temporal signals, while the geographic plot shows location-based variation in predicted demand.



Temporal signals were stronger than pure spatial signals. Color shows predicted demand; overlapping points indicate businesses in similar geographic areas.

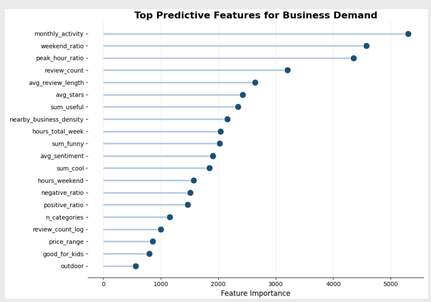
MODEL BENCHMARKING 7

Algorithm	RMSE	MAE	R^2
LightGBM (Tuned)	**0.3849**	**0.2948**	**94.50%**
XGBoost (Tuned)	0.3967	0.3018	94.16%
LightGBM	0.4096	0.3170	93.77%
XGBoost	0.4151	0.3206	93.60%
Random Forest	0.4383	0.3334	92.87%
Linear Regression	0.7466	0.5739	79.31%

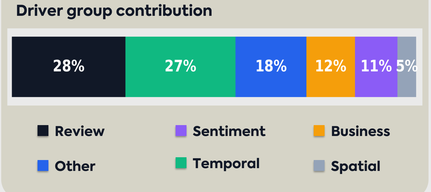


- Best regression model: LightGBM (Tuned).
- Reported metrics are from Notebook 04 loaded outputs.

BUSINESS DRIVER SUMMARY 8



Driver group contribution



- Review Activity and Temporal features account for 55% of grouped driver contribution.

KEY FINDINGS & CONCLUSION 9

Main Findings

- Yelp check-in activity can be used as a demand proxy.
- Tuned LightGBM produced the strongest regression result.
- SHAP identified temporal activity and review engagement as the clearest demand drivers.

RQ & hypothesis alignment

RQ	Hypothesis	Evidence	Status
RQ1	H1	Strong prediction performance	Supported
RQ2	H4	Temporal/spatial patterns	Supported
RQ3	H2	LightGBM outperformed baseline	Supported
RQ4	H3	SHAP identified key drivers	Supported

Conclusion

- The results support the proposal's objective and refine the location-based framing: Yelp data can estimate check-in-based demand, while temporal activity and review engagement are stronger demand signals than location alone.

IMPLICATIONS & LIMITATIONS 10

Strategic implications

- **Staffing & hours:** Use temporal signals for staffing and operating-hour planning.
- **Customer attention**: Use review activity as a demand signal, not only a satisfaction signal.
- **Location screening**: Combine nearby density and predicted demand for location evaluation.

Limitations & future work

- Check-ins are a demand proxy, not revenue or POS transactions.
- Findings are model-based, not causal proof.
- Future work: richer sentiment models, CatBoost, and external validation.

Primary data source: Yelp Open Dataset, April 2024 version.

Poster source: 03_Modeling_Phase_Rev04 + 04_Driver_Analysis_and_Interpretation_Rev01